Statistics 280 Winter 2025 Midterm 1

Name (Please print legibly)	UMID Number

- Please answer all questions in the space provided and indicate where in the exam you have written additional work.
- **Solutions without justification will be marked as wrong for all questions except 5d. Show all work. A calculator command does not count as justification**.
- Express answers as fractions or as decimals rounded to 2 decimal places.
- The last page is a table of values from the normal distribution.

Academic Integrity

Please read and sign the following statement regarding academic integrity to indicate that you have read, understand, and comply with the statement.

I have never given nor received unauthorized aid on this examination, nor have I concealed any violations of the academic integrity policy as stated by the College of Literature, Science, and the Arts. I have not done anything that may be understood as any action or attempted action that may result in creating an unfair academic advantage for oneself or an unfair academic advantage or disadvantage for any other member of members of the academic community. I understand that if I as suspected of academic misconduct, the incident may be reported to the Office of the Assistant Dean.

Signature: _____

1.(30 points) The Public Health Service studied the effects of smoking on health, in a large sample of representative households. For men and for women in each age group, those who had never smoked were on average somewhat healthier than the current smokers, but the current smokers were on average much healthier than those who had recently stopped smoking.

a) Why did they study men and women and the different age groups separately?

b) The lesson seems to be that you shouldn't start smoking, but once you've started, don't stop. Comment briefly.

c) Suppose that you want to control for several binary (two categories) variables in your study of smoking, and you want each comparison to consist of at least 15 smokers and 15 non-smokers. What is the maximum number of variables that you can control for if your total sample size is 1,000.

d) Is this a randomized controlled experiment or an observational study? Explain briefly.

2.(30 points) NFL quarterbacks have the following completion percentages on average by distance thrown: short (5 yards or less) 80.5%, mid (6-15 yards) 65%, deep (16-25 yards) 52%, bomb (26+ yards) 33.6%. An average NFL quarterback throws short roughly 45% of the time, mid 34% of the time, deep 15% of the time, and a bomb 6% of the time.

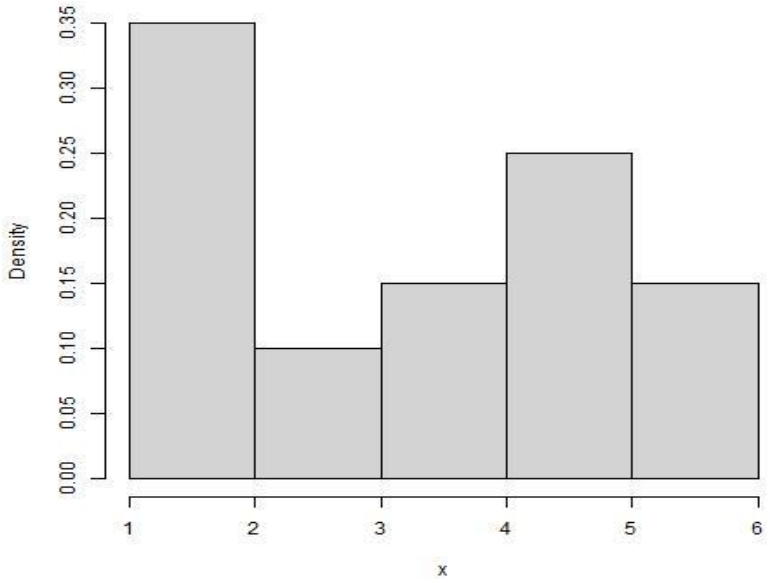
a) Estimate the completion percentage of an average NFL quarterback.

b) You are an agent representing a quarterback with a completion percentage of 60%. You claim that your quarterback's completion percentage is lower than the average in part a) because he throws deep more often that a typical quarterback. He throws 30% of throws short, 45% of throws mid, 20% of throws deep, and 5% bombs. Is your quarterback more accurate than an average quarterback who throws with the same distribution as your quarterback?

c) It is common to summarize a list of numbers with the **5 number summary**: the minimum, 25th percentile, median, 75th percentile, and maximum. Suppose that the 5 number summary of a list of 6 numbers is 1, 4, 5, 8 and 10, based on the percentile definition in class, and that the mean of the list is 35/6. What is the list of numbers?

d) All numbers in a given list are between 0 and 10. Is it possible for the standard deviation of the list to be greater than 10? If so, then given an example of such a list, and if not, then explain why not.

3. (35 points) a. Estimate the median, mean, and interquartile range for the values from the histogram below, assuming that that values are evenly distributed within each block.



b. What number would you use to summarize the histogram if you wanted to minimize the sum of the absolute values of the differences between that value and the values in the list represented by the histogram?

c. What number would you use to summarize the histogram if you wanted to minimize the sum of the squared differences between that value and the values in the list represented by the histogram?

4. (35 points) Suppose that the heights of 5 year old girls average 43 inches with a standard deviation of 3 inches.

a. Estimate the percentage of 5 year old girls who are between 42 and 47 inches tall, assuming that their heights are approximately normally distributed.

b. Estimate the percentage of 5 year old girls who are at least 48 inches tall, assuming that their heights are normally distributed.

c. Estimate the 15th percentile for height, assuming that heights are normally distributed.

d. What is the maximum possible percentage of 5 year old girls that are at least 48 inches tall? **Do not assume that heights are normally distributed.**

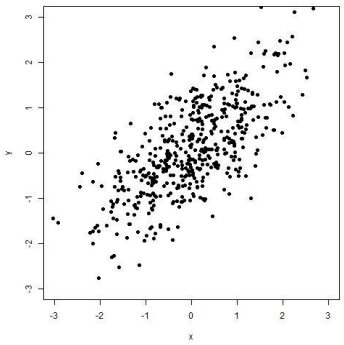
e. What is the maximum possible percentage of 5 year old girls with heights that are more than 47.5 inches or less than 38.5 inches? **Do not assume that heights are normally distributed.**

5. (30 points) a) Suppose that the correlation between height and weight for 100 5 year old girls is 0.2, that the average and standard deviation for height are 43 inches and 3 inches, respectively, and that the average and standard deviation for weight are 40 pounds and 10 pounds, respectively. Find the sum of the products of height and weight for these 100 5 year old girls.

b) Suppose that the correlation between height and weight for 100 10 year old girls is 0.2, that the average and standard deviation for height are 55 inches and 3 inches, respectively, and that the average and standard deviation for weight are 70 pounds and 15 pounds, respectively. Find the sum of the products of height and weight for these 100 5 year old girls.

c) Suppose that we combine the height and weight data in parts a) and b). The means for height and weight in the combined dataset are 49 inches and 55 pounds, respectively. The standard deviations for height and weight in the combined dataset are 6.7 inches and 19.7 pounds, respectively. Find the correlation between height and weight in the combined dataset.

d) Is the correlation between x and y in the scatter plot below 0, 0.7, or -0.5?



6. (40 points) Suppose that house cats have weights that average 10 pounds with a standard deviation of 2 pounds and lengths that average 28 inches with a standard deviation of 3 inches. Suppose further that the correlation between height and weight for these house cats is 0.5. Assume that the scatter plot for height and weight for these house cats is football-shaped. Note that questions continue onto the next page.

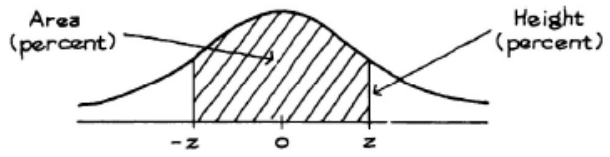
6a) What would be the predicted length for a 13 pound house cat?

6b) Give an equation for the predicted length y in terms of a given weight x.

6c) Estimate the interquartile range for length for 13 pound house cats? (Hint: what percentiles and standard units correspond to the endpoints of the interquartile range?)

6d) What length would you predict for a house cat whose weight was in the 25th percentile?

Tables



A NORMAL TABLE

z	*Height*	*Area*	*z*	*Height*	*Area*	*z*	*Height*	*Area*
0.00	39.89	0	1.50	12.95	86.64	3.00	0.443	99.730
0.05	39.84	3.99	1.55	12.00	87.89	3.05	0.381	99.771
0.10	39.69	7.97	1.60	11.09	89.04	3.10	0.327	99.806
0.15	39.45	11.92	1.65	10.23	90.11	3.15	0.279	99.837
0.20	39.10	15.85	1.70	9.40	91.09	3.20	0.238	99.863
0.25	38.67	19.74	1.75	8.63	91.99	3.25	0.203	99.885
0.30	38.14	23.58	1.80	7.90	92.81	3.30	0.172	99.903
0.35	37.52	27.37	1.85	7.21	93.57	3.35	0.146	99.919
0.40	36.83	31.08	1.90	6.56	94.26	3.40	0.123	99.933
0.45	36.05	34.73	1.95	5.96	94.88	3.45	0.104	99.944
0.50	35.21	38.29	2.00	5.40	95.45	3.50	0.087	99.953
0.55	34.29	41.77	2.05	4.88	95.96	3.55	0.073	99.961
0.60	33.32	45.15	2.10	4.40	96.43	3.60	0.061	99.968
0.65	32.30	48.43	2.15	3.96	96.84	3.65	0.051	99.974
0.70	31.23	51.61	2.20	3.55	97.22	3.70	0.042	99.978
0.75	30.11	54.67	2.25	3.17	97.56	3.75	0.035	99.982
0.80	28.97	57.63	2.30	2.83	97.86	3.80	0.029	99.986
0.85	27.80	60.47	2.35	2.52	98.12	3.85	0.024	99.988
0.90	26.61	63.19	2.40	2.24	98.36	3.90	0.020	99.990
0.95	25.41	65.79	2.45	1.98	98.57	3.95	0.016	99.992
1.00	24.20	68.27	2.50	1.75	98.76	4.00	0.013	99.9937
1.05	22.99	70.63	2.55	1.54	98.92	4.05	0.011	99.9949
1.10	21.79	72.87	2.60	1.36	99.07	4.10	0.009	99.9959
1.15	20.59	74.99	2.65	1.19	99.20	4.15	0.007	99.9967
1.20	19.42	76.99	2.70	1.04	99.31	4.20	0.006	99.9973
1.25	18.26	78.87	2.75	0.91	99.40	4.25	0.005	99.9979
1.30	17.14	80.64	2.80	0.79	99.49	4.30	0.004	99.9983
1.35	16.04	82.30	2.85	9.69	99.56	4.35	0.003	99.9986
1.40	14.97	83.85	2.90	0.60	99.63	4.40	0.002	99.9989
1.45	13.94	85.29	2.95	0.51	99.68	4.45	0.002	99.9991